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Growth and characterization of nonlinear optical telluromolybdate $CoTeMoO_6$ single crystals

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ABSTRACT

Nonlinear optical (NLO) crystals $CoTeMoO_6$ have been grown from high temperature solution by means of spontaneous crystallization. These crystals have been characterized by temperature-dependent X-ray diffraction, infrared and electron absorption methods. Moreover, polarized Raman spectra have been measured at room temperature. These studies showed that the Mo^{6+} and Te^{4+} ions are in asymmetric coordination environment, and $CoTeMoO_6$ has the largest flexibility index among $MTeMoO_6$ family of telluro-molybdates (M=Zn, Mg, Cd, Mn, Co), making this compound attractive NLO material. Raman and IR studies allowed assigning the observed modes to respective motions of atoms in the unit cell. They have also showed evidence for spin–phonon coupling at low temperatures and suggested that $CoTeMoO_6$ is prospective material for solid-state Raman laser-frequency converters. Analysis of electron absorption data has shown that Co^{2+} ions are located in sites of weak crystal field with the crystal field parameters Dq and B equal to 548 cm^{-1} and 882 cm^{-1}, respectively.

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1. Introduction

Second-order nonlinear optical (NLO) crystals are of interest for possible applications in frequency shifting, sensing, optical communications, etc. Particularly interesting NLO crystals are compounds containing Mo^{6+} or W^{6+} and cations with nonbonded electron pairs such as Se^{4+} and Te^{4+} since these cations may exhibit the second-order Jahn–Teller (SOJT) effect, which leads to the noncentrosymmetric structure and strong NLO properties in some telluromolybdates, for instance β-$BaTeMo_2O_9$ [1–8]. One of promising family of telluromolybdates are $ATeMoO_6$ compounds (A=Mg, Zn, Mn, Co), which crystallize in noncentrosymmetric orthorhombic crystal system with space group $P2_12_12$ [9–13]. Among this family of compounds, single crystal growth was reported recently for Mn, Zn and Mg compounds and it has been shown that these crystals display a large second harmonic generation (SHG) response of approximately 0.7, 1.1 and 1.5 × KTP, respectively [11–13]. It is worth noting that single crystals growth was also reported for closely related $CdTeMoO_6$, which however crystallizes with space group $P\bar{4}2_1m$ [14]. Optical measurements also indicated that $ATeMoO_6$ crystals (A=Mg, Zn, Cd) are transparent from about 0.35 to 5.4 μm [12–14]. $MnTeMoO_6$ is also transparent in a broad wavelength range, except of the 0.47–0.52 μm range [11]. $CoTeMoO_6$ has also attracted significant interest but mainly due to its catalytic, structural and magnetic properties [9,10]. These studies were, however, limited to polycrystalline powders and growth as well as characterization of $CoTeMoO_6$ single crystals have not been reported yet.

In this paper, we report for the first time the bulk growth of $CoTeMoO_6$ single crystals by means of high temperature solution growth method. The obtained crystals are characterized by electron absorption and IR absorption spectroscopes in a broad temperature range from 295 to 5 K. We also report temperature-dependent single crystal X-ray diffraction and polarized Raman studies on $CoTeMoO_6$ single crystal at room temperature. We would like to emphasize that electron absorption, Raman and IR studies have not yet been reported for this family of NLO crystals. We assign the observed modes to respective vibrations of atoms in the unit cell and show evidence for spin–phonon coupling at low temperatures due to onset of magnetic order.

2. Experimental section

2.1. Growth of $CoTeMoO_6$ single crystals

$CoTeMo_2O_6$ single crystals were obtained from high temperature solution by means of spontaneous crystallization on a Pt wire

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immersed in the melt. As a solvent TeO_2–MoO_3 mixture (1:0.8 in molar ratio) was used. The molar composition of the solvent was chosen using the phase diagram given by Petrini and Bart [15]. The composition corresponds to eutectics and therefore has the lowest melting temperature of 526 °C. The concentration of $CoTeMo_2O_6$ in the solvent was 16.6 mol%. The starting melt was synthesized directly in a Pt crucible used in the experiments. To ensure the uniform composition the melt was mechanically stirred with use of a Pt stirrer for several hours. The crucible was placed in a two-zone furnace enabling crystallization under conditions of low temperature gradients. The material shows strong tendency to spontaneous crystallization in form of platelets, so in this phase of our experiments we used Pt wire as a seed. Fig. 1 shows the bunch of as-grown blue $CoTeMoO_6$ platelets (grid 5 mm) obtained after two weeks of slow cooling of the melt at a rate of 0.01 °C h^{-1}.

2.2. Crystallographic structure determination

Single-crystal diffraction data were collected at 298 and 150 K on a 4-circle diffractometer KM4CCD (Oxford Diffraction) using a graphite monochromated Mo*K*α radiation (λ=0.71073 Å). For the low temperature measurements, the crystal temperature was maintained using an open flow nitrogen cryosystem (Oxford Cryosystem). Thermal evolution of the lattice parameters was measured from 100 to 300 K with a step 40 K. Lattice parameters were obtained by a least square fit to 142 reflections. The CrysAlis RED software version 1.170.32 (Oxford Diffraction) was used for data processing [16]. An empirical absorption correction was applied using spherical harmonics implemented in Scale3 Abspack scaling algorithm. The structure was refined by full-matrix least-squares method by means of SHELX-97 program package [17].

2.3. Optical transmission

The absorption spectra were recorded on the Cary 5E spectrophotometer. The spectral resolution was 0.5 nm in the visible and 1.0 nm in the near infrared region. The thickness of the crystal was 0.15 mm.

2.4. Raman and IR studies

Infrared spectra of powdered sample were measured with the Biorad 575C FT-IR spectrometer in KBr pellet for the 4000–400 cm^{-1} region and in Nujol suspension for the 500–50 cm^{-1} region. Polarized Raman spectra were measured using the Renishaw InVia Raman spectrometer equipped with confocal DM 2500 Leica optical microscope, a thermoelectrically cooled CCD as a detector and a diode laser operating at 830 nm. The spectral resolution was 2 cm^{-1}. The orientation of the crystallographic axes of the studied crystal was done using X-ray diffraction method.

Fig. 1. The bunch of as-grown $CoTeMoO_6$ platelets (grid 5 mm).

3. Results and discussion

3.1. Single crystal X-ray diffraction

The crystal structure of $CoTeMoO_6$ has been solved ab initio by Laligant and later refined by Doi et al. using the Rietveld method [9,10]. To reconfirm the crystal structure, we have collected single-crystal X-ray diffraction data at two temperatures. The details of data collection, data reduction, and crystallographic data are summarized in Tables S1–S3. The compound crystallizes in the $P2_12_12$ space group and tha lattice parameters are as follow: a=5.2427 Å, b=5.0621 Å, c=8.8464 Å, Z=2, V=234.78 $Å^3$ (see Table S1). There obtained parameters and bond lengths in our work are very similar to those reported by Doi et al. [10]. Co^{2+} ions have distorted octahedral coordination and the octahdra are connected by sharing the corner oxygen O3 (Fig. 2). The Mo^{6+} and Te^{4+} ions have tetrahedral coordination and they connect to the CoO_6 octahedra forming layers in the *ab*-plane (Fig. 2). The Mo–O bond distance in the MoO_4 tetrahedron range from 1.698 to 1.859 Å. The Te–O bond distances are 1.891 and 2.081 Å and the strongly asymmetric coordination of Te atoms can be attributed to SOJT effect.

The crystal structure of $CoTeMoO_6$ does not change at low temperatures. The temperature dependence of the lattice parameters show that they decrease upon cooling (see Fig. S1). This decrease is the fastest for the parameter c and the temperature variation of this parameter can be approximated by the equation $c=c_0+aT+bT^2$ with the intercept c_0=8.8152 Å and coefficients $a=-8.9286\times10^{-7}$ Å K^{-1} and b=3.6830×10^{-7} Å K^{-2}. The unit cell volume can also be approximated by the equation $V=V_0+aT+bT^2$ with the intercept V_0=233.68 $Å^3$ and coefficients $a=-0.00164$ $Å^3$ K^{-1} and b=2.6786 $\times10^{-5}$ $Å^3$ K^{-2}. The parameters a and b have similar temperature dependence and they decrease linearly upon cooling with the intercepts a_0=5.05621 and b_0=5.2279 Å, and temperature coefficient 5.6429 and 5.7143 Å K^{-1}, respectively. More pronounced temperature dependence of the parameter c could be attributed to the layered structure of $CoTeMoO_6$ and weak intermolecular interactions between layers in the c direction.

The structure of $CoTeMoO_6$ is very similar to the isostructural Zn, Mn and Mg compounds, i.e., all these crystals are composed of very similar asymmetric TeO_4 and MoO_4 groups as well as MO_6 (M=Zn, Mg, Mn, Co) octahedra. Thus similar as the other compounds, $CoTeMoO_6$ is also expected to exhibit significant NLO properties. It has been argued that strong SHG response of these crystals can be attributed to contribution of NLO-active TeO_4, MoO_4 and MO_6 units [11–13]. However, recent first-principles calculations of SHG coefficients reported for MTeMoO$_6$ (M=Mg, Zn, Cd) revised this opinion, that is, they showed that divalent cations have little contribution to the NLO response [18]. More interestingly, although the TeO_4 groups are usually beneficial to the generation of a large NLO response due to the existence of lone-pair electrons, these groups contribute weakly to SHG effects in MTeMoO$_6$ crystals [18]. Thus more than 90% of the overall SHG coefficients stem from the MoO_4 groups [18].

Jiang et al. also proposed that the SHG effect in a crystal will increase with increased "flexibility" of the chemical bonds, provided the resulting second-order susceptibility can be additively superposed

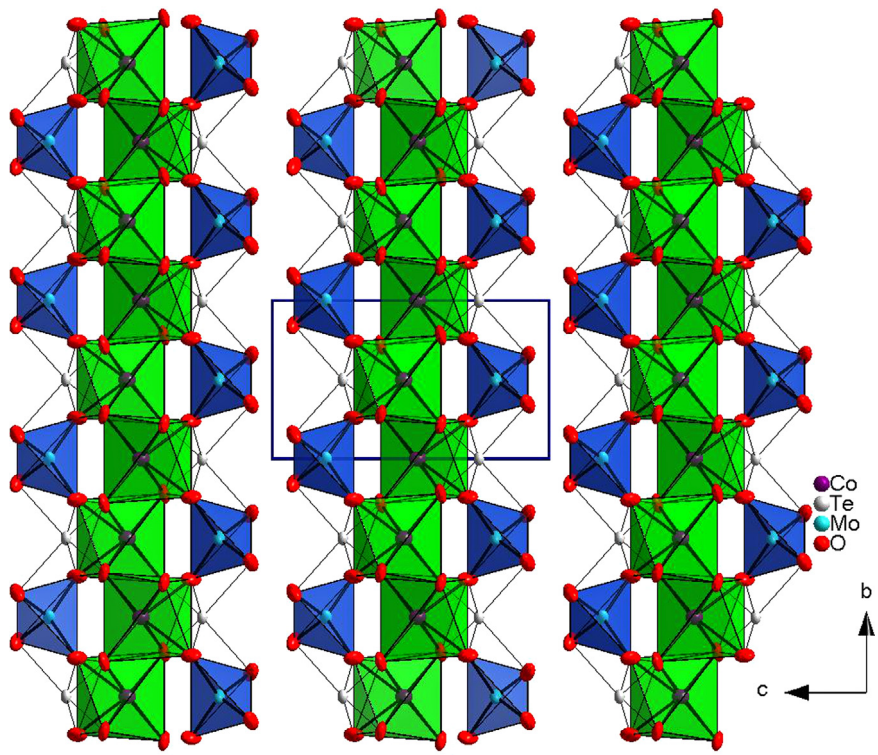

Fig. 2. View of the crystal structure along the *a*-axis.

[18]. In the proposed model, the flexibility index F can be estimated using the following equation:

$$F = \frac{\exp\left[(R_0 - R_a)/B\right]}{\left(\sqrt{C_a} + \sqrt{C_b}\right)^2 / R_a^2} \quad (1)$$

where R_a denotes the average bond length of the group contributing to the SHG effect, R_0 is the ideal bond length when the atom contributes exactly unit valence to the central atom [19], B is an empirical constant, typically 0.37 [20], and C_a (C_b) for Mo and O are 6 [18]. Using Eq. (1) we have calculated the F indices for the studied here $CoTeMoO_6$ crystal and reported previously $MnTeMoO_6$ crystal. For the comparison sake, we also list the values calculated previously for Zn, Mg and Cd analogues in Ref. 18 (see Table 1). Table 1 shows that $CoTeMoO_6$ has the smallest value of R_a and the largest values of the binding force as well as the F index among all known MTeMoO$_6$ compounds. On the other hand, the F index of $MnTeMoO_6$ is significantly smaller. Here it is important to add that for the orthorhombic point group 222, d_{14}, d_{25} and d_{36} are the only non-zero coefficients of the SHG tensor and d_{14} is the strongest component in the family of ATeMoO$_6$ crystals [18]. Jiang et al. showed that although the proposed flexibility model takes into account only the induced dipole moment of a microscopic group that has the strongest contribution to the SHG effect and ignores other effects, the largest SHG coefficient in a family of isostructural compounds is proportional to the F index [18]. Our data suggest, therefore, that $CoTeMoO_6$ crystal is expected to exhibit stronger SHG response than its manganese analogue. Although our present data do not give information on phase matching behavior of d_{14} in $CoTeMoO_6$, type-I phase matchable behavior at 1064 nm was found for isostructural $ZnTeMoO_6$ and $MnTeMoO_6$ as well as closely related $CdTeMoO_6$ [11,13,14]. It is therefore very likely that $CoTeMoO_6$ will also show phase matching behavior.

Table 1
Values of the average Mo–O bond length (R_a), bond valence $\exp[(R_0-R_a)/B]$, binding force of the bond electrons to the cores ($(\sqrt{C_a}+\sqrt{C_b})^2/R_a^2$) and the flexibility index for all known MTeMoO$_6$ compounds.

	R_a (Mo–O) (Å)	$\exp[(R_0-R_a)/B]$	$\left(\sqrt{C_a}+\sqrt{C_b}\right)^2/R_a^2$	F
$CoTeMoO_6$	1.7785	1.4152	7.5876	0.1865
$MnTeMoO_6$	1.7848	1.3913	7.5341	0.1847
$MgTeMoO_6$	1.782	1.402	7.558	0.186
$CdTeMoO_6$	1.783	1.398	7.549	0.185
$ZnTeMoO_6$	1.786	1.387	7.524	0.184

3.2. Vibrational properties of the $CoTeMoO_6$ single crystal at room temperature

For the $P2_12_12$ structure of $CoTeMoO_6$ group theory predicts $12A+12B_1+15B_2+15B_2$ Brillouin zone center modes. Co^{2+}, Te^{4+} and Mo^{6+} ions give rise to $3A+3B_1+6B_2+6B_3$ translational modes. From these translational modes, $B_1+B_2+B_3$ modes belong to the acoustic branches. Vibrations of the oxygen ions contribute to $9A+9B_1+9B_2+9B_3$ Brillouin zone center modes. The A modes are Raman-active whereas the B_1, B_2 and B_3 modes are both Raman- and IR-active.

The structure of $CoTeMoO_6$ consists of MoO_4, CoO_6 and TeO_4 polyhedra. In the MoO_4 tetrahedra, the Mo–O1 bonds are very short (1.698 Å) since the O1 atoms do not connect to other ions except for Mo^{6+}. Vibrations of this oxygen atom will be denoted as stretching (ν) and bending (δ) vibrations of short Mo–O bonds. There should be $A+B_1+B_2+B_3$ and $2A+2B_1+2B_2+2B_3$ stretching and bending modes of these short Mo–O bonds. The two remaining O2 oxygen ions in the MoO_4 tetrahedron are shared with the TeO_4 polyhedron. As a result, the corresponding Mo–O2 bonds are longer, i.e., 1.859 Å, and the Te–O2 bond distances are also relatively long, i.e. 2.082 Å. We will describe vibrations of these

oxygen ions as vibrations of Mo–O–Te bridges and there should be $3A+3B_1+3B_2+3B_3$ such modes. The O3 oxygen ions form short Te–O bond distances of 1.892 Å and we will classified vibrations of these ions as stretching and bending vibrations of short Te–O bonds. There should be $A+B_1+B_2+B_3$ and $2A+2B_1+2B_2+2B_3$ stretching and bending modes of these short Te–O bonds.

Figs. S2 and S3 present IR spectra of powdered $CoTeMoO_6$ at different temperatures in the mid-IR region and room-temperature spectrum in the far-IR region, respectively. Room temperature Raman spectra of the $CoTeMoO_6$ single crystal, measured in different polarization configurations, are shown in Fig. 3. All observed Raman and IR wavenumbers are summarized in Table S4.

Let us now discuss origin of the observed modes. Fig. 3, S2 and S3, and Table S4 show that there is a clear separation between the higher wavenumber modes (above 610 cm^{-1}) and the low wavenumber modes (below 490 cm^{-1}). The former range can be attributed to stretching modes and the latter to bending and lattice modes. Based on literature data for molybdates, tellurates containing TeO_4 units and telluromolybdates, we may assign the 890–960 cm^{-1} bands to stretching vibrations of the short Mo–O bonds, whereas the bands in the range and 610–790 cm^{-1} correspond to stretching vibrations of Mo–O bonds and Mo–O–Te bridges [6,7,21–24]. In a similar way, we may propose assignment of the 250–480 cm^{-1} bands to bending modes of the Mo–O and Te–O bonds as well as Mo–O–Te bridges.

According to literature data, translations of molybdenum and tellurium atoms contribute to lattice modes observed below 200 cm^{-1} [6,7,21–24]. Translations of Co^{2+} are usually observed in the same region as Mn, Ni, Fe, i.e. in the 300–150 cm^{-1} [25,26]. We suppose, therefore, that the bands near 240–260 cm^{-1} have large contribution from Co^{2+} vibration whereas those below 200 cm^{-1} can be assigned mainly to vibrations of the heavy molybdenum and tellurium atoms.

Raman data allow us also to estimate whether this material can find application as a Raman laser converter (shifter) based on stimulated Raman scattering (SRS) effect. The amplification of a Raman band in SRS process is expected to be most efficient for modes exhibiting large spontaneous Raman scattering cross-section and narrow bandwidth [27]. Our Raman spectra show that the highest intensity Raman bands are observed in $b(cc)\bar{b}$ polarization at 102, 115, 147, 379, 707 and 951 cm^{-1}. The narrowest bands are those at 147 and 951 cm^{-1} (bandwidth 4.4 and 3.6 cm^{-1}, respectively). Therefore, the most intense lasing lines of the SRS spectra are expected to appear for the 147 and 951 cm^{-1} modes. Large Raman shift of the latter mode is beneficial since it would allow large tunability of the Raman shifters based on this material.

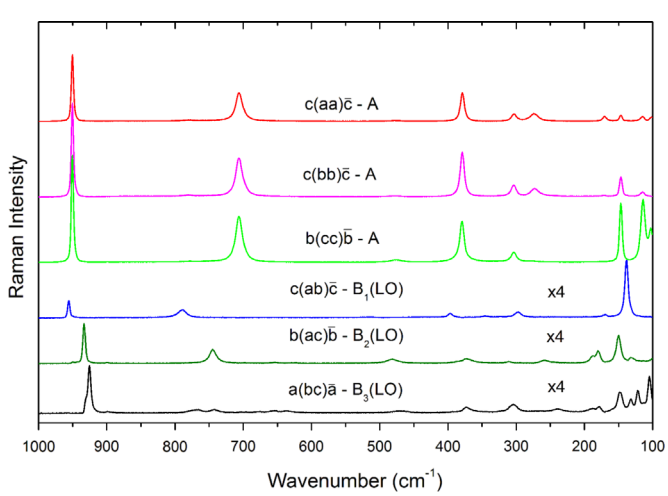

Fig. 3. Polarized Raman spectra of $CoTeMoO_6$ single crystal recorded at 295 K.

3.3. Temperature-dependent vibrational properties of the $CoTeMoO_6$ single crystal

The IR spectra presented in Fig. S2 show relatively weak changes and no splitting of bands or appearance of new bands is apparent that could be indicative of a structural phase transition. To observe the temperature dependence of the IR bands more clearly, we present in Fig. 4 the temperature dependence of IR wavenumbers for $CoTeMoO_6$. There might be four contributions to change in wavenumber of a phonon i [28]:

$$\Delta\omega_i(T) = (\Delta\omega_i)_{latt} + (\Delta\omega_i)_{anh} + (\Delta\omega_i)_{ren} + (\Delta\omega_i)_{s-ph} \quad (2)$$

The first contribution reflects the change of the ionic binding energies due to lattice expansion, second is the intrinsic anharmonic contribution, third arises from a renormalization of the electronic states and the last contribution describes the spin–phonon coupling [28]. We neglect the third term since $CoTeMoO_6$ is an isolator. The fourth term becomes important only at those temperatures where magnetic ordering exists [26,28,29]. Since $CoTeMoO_6$ shows magnetic transition at 24.4 K [10], the observed by us temperature dependence of $CoTeMoO_6$ phonon modes results mainly from phonon–phonon anharmonic effects with additional contribution at low temperatures due to spin–phonon interactions.

Fig. 4 shows that with decreasing temperature all modes exhibit usual hardening due to phonon–phonon anharmonic interactions. The modes above 890 cm^{-1} exhibit very weak increase of

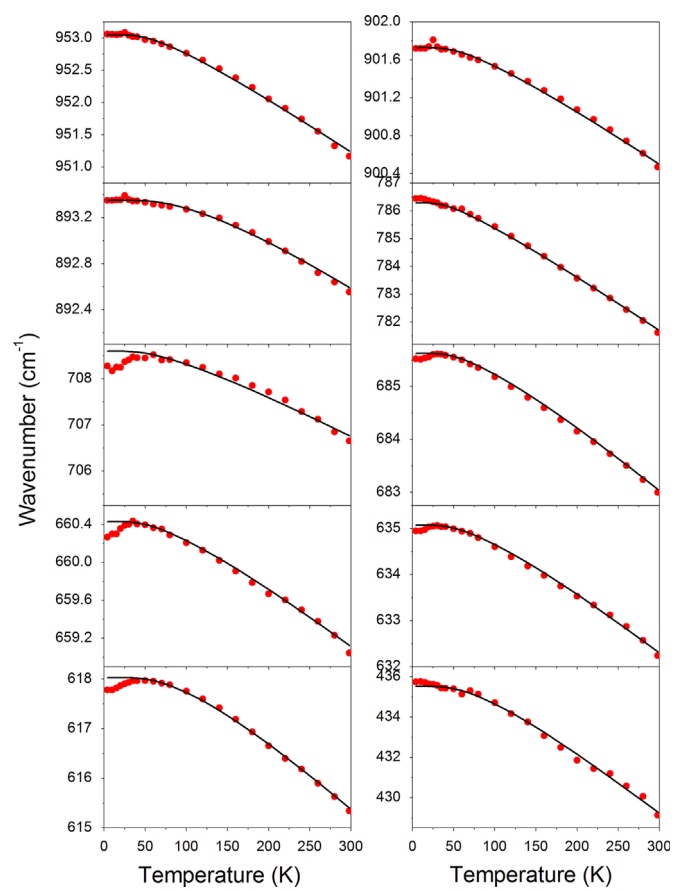

Fig. 4. Temperature dependences of IR wavenumbers of $CoTeMoO_6$. Solid lines correspond to the expected temperature dependence of the phonon wavenumbers due to phonon-phonon interactions.

wavenumbers near 25 K. However, the modes in the 600–790 cm^{-1} range exhibit more clear anomalies, i.e., the 782 cm^{-1} mode (600–710 cm^{-1} modes) exhibits upshift (downshift) by 0.25 cm^{-1} (up to 0.3 cm^{-1}) below 25 K (see Fig. 4). These additional weak anomalies can be attributed to spin–phonon coupling. Theoretical considerations showed that amount of the additional wavenumber shift is proportional to the spin correlation function $\langle S_i S_j \rangle$ for the spins localized in the ith and jth magnetic ion sites [28]:

$$\Delta\omega_i \equiv \lambda \langle S_i S_j \rangle \tag{3}$$

where the spin–phonon coupling constant λ is different for different phonons and can have either a positive or a negative sign. In general, significant changes are expected for those phonons, which involve motions that modulate spin exchange coupling. Doi et al. showed that the magnetic order is associated with a canted antiferromagnetic ordering of Co^{2+} ions [10]. The antiferromagnetic component of the magnetic moment is along the b-axis but there is also an order of magnitude smaller ferromagnetic component along the a-axis [10]. Since in the studied here $CoTeMoO_6$ the magnetic exchange occurs through the O2 and O3 oxygen ions, one may expect to observe some anomalies for the modes involving vibrations of these oxygens and no or very weak anomalies for the O1 ions that are connected only to Mo^{6+} ions. Our IR data are consistent with this assumption, showing nearly no anomalies for the modes above 890 cm^{-1}, which correspond to vibrations of the short Mo–O1 bonds, and clear anomalies for the modes in the 600–800 cm^{-1} range, which correspond to vibrations of Te–O3 bonds and Mo–O2–Te bridges.

3.4. Optical Properties of the $CoTeMoO_6$ single crystal

Fig. 5 shows the transmittance spectra of $CoTeMoO_6$ at room temperature. The UV absorption edge is located at 385 nm and the spectra show presence of a few bands in the UV–vis and near-IR region. Presence of these bands would affect application of $CoTeMoO_6$ as a NLO material, especially in the UV–vis region.

The optical properties of d^n ions are interpreted by means of a crystal field energy diagram originating from the free-ion LS-terms of the $3d^n$ configuration. The free-ion energies are functions of inter-electron interaction Racah parameters B and C. The interaction of the $3d$ electrons with the octahedral lattice surroundings is characterized by the parameter Dq describing strength of the octahedral crystal field. The octahedral crystal field acting on Co^{2+} ion splits the sevenfold degenerated ground term ^4F of the free ion into a triplet orbitals $^4T_{1g}$, $^4T_{2g}$ and single $^4A_{2g}$ orbital. The spin allowed transitions are expected from the ground level $^4T_{1g}$ to the excited $^4T_{2g}$ and $^4A_{2g}$ states as well as to a higher $^4T_{1g}$ state from the unsplit ^4P term of the free ion [30]. The observed optical absorption transitions allow to evaluate these crystal-field parameters. The energies of spin-allowed transitions in octahedral and tetrahedral d^n (n=2, 3, 7, 8) systems, depend on the parameters $10Dq$ and B only. In particular, these expressions are independent of the interelectronic repulsion parameter C. According to the crystal-field theory the energy matrix of $3d^7$ ions in octahedral crystal-field with electron configuration $t_{2g}^5 e_g^2$ (high-spin case) yields the energy levels for d^7 (Co^{2+}) ions in following form [31]:

$$E(^4T_2(^4F)) = 2Dq \tag{4}$$

$$E(^4A_2(^4F)) = 12Dq \tag{5}$$

$$E(^4T_1(^4F)) = -7.5B - 3Dq - 1/2(225B^2 + 100Dq^2 + 180Dq)^{1/2} \tag{6}$$

$$E(^4T_1(^4P))0 = -7.5B - 3Dq + 1/2(225B^2 + 100Dq^2 + 180Dq)^{1/2} \tag{7}$$

The parameter of the octahedral crystal-field Dq and the Racah parameter B are given by the following equations:

$$E_1 = E(^4A_2) - E(^4T_2(^4F)) = 10Dq \tag{8}$$

$$E_2 = E(^4T_1(^4P)) - E(^4T_1(^4F)) = (225B^2 + 100Dq^2 + 180DqB)^{1/2} \tag{9}$$

Polarized absorption spectra of $CoTeMoO_6$ measured at room temperature and the nonpolarized absorption spectrum at 5 K are presented in Fig. 6. The spectra are characterized by three main regions of absorption, two in the near IR region around 1640–1560 nm (6100–6400 cm^{-1}) and 862–826 nm (11600–12100 cm^{-1}), and one in the visible region 585–583 nm (17100–17150 cm^{-1}) (see Fig. 6). These bands – in a first approximation – can be ascribed to spin-allowed d–d transitions of Co^{2+} ions in octahedral crystal field of oxygen ligands [32–34]. The $^4T_{1g}(F)$–$^4T_{2g}(F)$ transition of the room temperature spectra corresponds to the absorption bands with absorbance maximum at 1623 nm (6160 cm^{-1}), the $^4T_{1g}(F)$–$^4A_{2g}(F)$ transition involves broadband absorption at 859 nm (11640 cm^{-1}) and the third $^4T_{1g}(F)$–$^4T_{1g}(P)$ transition gives rise to the prominent band in the visible region at 585 nm (17100 cm^{-1}). Weak bands or shoulders observed in the contour of this band at 16260, 18740, 19920, 20550, 21400, 22580 and 23280 cm^{-1} are caused by spin-forbidden transitions from the quartet ground state to higher doublet states (derived mainly from the ^2G and ^2H terms of the free ion).

The positions of the three maxima of the Co^{2+} band exhibit shifts to higher energies with temperature lowering down to 5 K. For the infrared bands the absorbance maxima shift from 6160 to

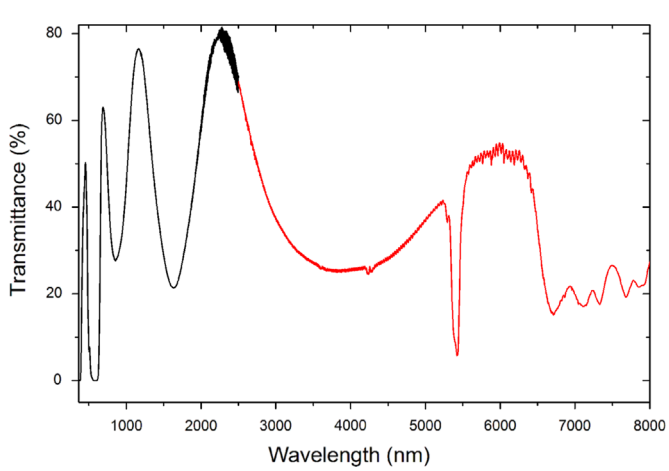

Fig. 5. UV–vis and IR transmittance spectra of $CoTeMoO_6$ single crystals at room temperature.

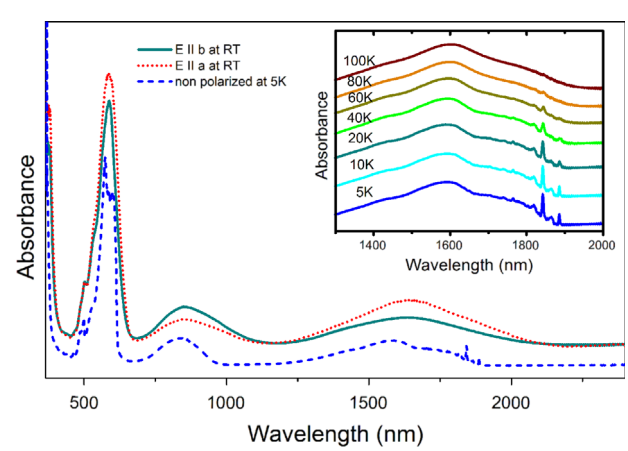

Fig. 6. Polarized absorption spectra of $CoTeMoO_6$ single crystals at T=298 K and nonpolarized spectrum at 5 K. Polarization directions: $E \parallel x$ ($\parallel a$) and $E \parallel y$ ($\parallel b$). Inset shows the IR range of absorption spectra with lowering temperature.

6380 cm^{-1} and from 11640 to 12080 cm^{-1}, respectively. The maximum of the visible absorption bands exhibits much smaller shift, i.e., from 17100 to 17150 cm^{-1}. The lowering of temperature leads also to changes in the contours of the $^4T_{1g}(F)$–$^4T_{2g}(F)$ and $^4T_{1g}(F)$–$^4T_{2g}(P)$ bands. The former band shows additional components at 5 K: one broad shoulder at 5730 cm^{-1} with group of narrow lines in the region 5295–5750 cm^{-1} and the second broad shoulder at 7030 cm^{-1}. The profile of the visible band splits into two clear components at 16620 and 17405 cm^{-1} and a weak component at 17060 cm^{-1}. The observed spectral fine structure and the polarization of the absorption bands can be attributed to presence of the cobalt ions in the non-regular coordination environment (C_2 point symmetry of the Co^{2+} sites), which lifts degeneracies of the $^4T_{2g}$ states. Similarly, bands corresponding to the spin-forbidden transitions in the range 16000–24000 cm^{-1} also become more pronounced and those above 18000 cm^{-1} exhibit weak shift to higher energies. The schematic energy level diagram for quartet and some doubled states of Co^{2+} ion in crystal fields of octahedral and distorted octahedral symmetry (C_2 distortion) was constructed and presented in Fig. 7.

For energy values E_1 and E_2 (5480 cm^{-1} and 11640 cm^{-1}) calculated from transition energies at room temperature, the crystal field parameters Dq and B for Co^{2+} ion in the $CoTeMoO_6$ crystal were evaluated under near octahedral symmetry assumption. They are equal to 548 cm^{-1} and 882 cm^{-1}, respectively. The strength of the crystal field described by factor Dq/B equal 0.62 indicates weak crystal field of the oxygen ligands acting on Co^{2+} ion. The Racah parameter B is reduced compared to the free ion value (B free ion = 1038 cm^{-1}) due to covalency effects. For the calculated value Dq/B the observed transition energies to the $^4T_{2g}(F)$ and $^4A_{2g}(F)$ excited states are somewhat higher than the expected energies from the Tanabe -Sugano diagram for octahedral d^7 ion. Energy of the $^4T_{1g}(F)$—$^4T_{2g}(P)$ transition is in good agreement with that predicted by the Tanabe–Sugano diagram. Assumption of near octahedral symmetry allows us also to estimate change of the crystal field strength upon cooling. Our results show that somewhat higher values of Dq parameter (570 cm^{-1}) and crystal strength parameter Dq/B (0.65) are observed for $CoTeMoO_6$ at 5 K.

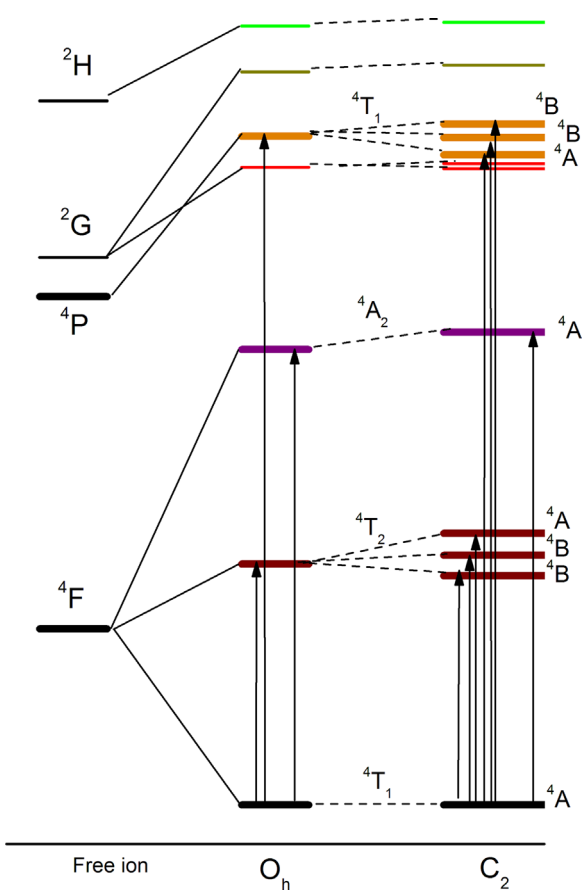

Fig. 7. Schematic crystal field splitting diagram $O_h - C_2$ with observed spin-allowed electron transitions for Co^{2+} ion.

It is worth to mention that when temperature decreased below 100 K, a group of sharp lines appeared in the 5295–5750 cm^{-1} region (see inset to Fig. 6). These sharp lines cannot be explained by the possible transitions deduced from the energy level diagram for Co^{2+} ion at C_2 sites in $CoTeMoO_6$. However, they can be explained in the same way as proposed by Ferguson et al., who assigned the sharp lines of Co^{2+} in octahedral coordination assuming lifting the degeneracy of the $^4T_1$ and $^4T_2$ triplets into one Γ_6, two Γ_8, and one Γ_7 spin–orbit components [35]. Thus the sharp lines were attributed to the transitions from ground Γ_6 component to the excited Γ_6, Γ_8, Γ_8 and Γ_7 spin–orbit components of the $^4T_2$ (O_h) state [35]. We may most likely also assign these lines to the transitions from the ground spin–orbit component of the $^4T_1$ state to spin–orbit components of the $^4T_2$ state but lower crystal field symmetry (C_2) in $CoTeMoO_6$ produces further splitting of the $^4T_1$ and $^4T_2$ triplets, giving rise to six Kramers doublets for each state [36].

4. Conclusions

We have successfully grown $CoTeMoO_6$ single crystals from high temperature solution by means of spontaneous crystallization. This compound crystallizes in the space group $P2_12_12$, and is isostructural to MTeMoO$_6$ compounds with M=Zn, Mg, Mn. Crystallographic data reveal that $CoTeMoO_6$ has the largest flexibility index among this family of telluromolybdates and is expected to show strong NLO properties. Analysis of electron absorption spectra down to 5 K has allowed us to estimate the crystal field parameters Dq and B for Co^{2+} ion in the $CoTeMoO_6$ crystal as equal to 548 cm^{-1} and 882 cm^{-1}, respectively. Values of these parameters point to weak crystal field of the oxygen ligands acting on Co^{2+} ion. Spectroscopic data has also revealed presence of many strong Raman bands related to totally symmetric modes. Among these modes, the most intense SRS effect is expected for the 147 and 951 cm^{-1} modes. Thus $CoTeMoO_6$ is prospective material for solid-state Raman laser-frequency converters.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.jssc.2014.08.023.

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